

SOFTWARE REQUIREMENTS SPECIFICATION

Smart Warehouse Management System (WMS)
with Intelligent Logistics Optimization

Scope Focus: Core Logistics Workflows & Practical Business Intelligence Solutions

TABLE OF CONTENTS

1. Introduction & Executive Summary

- 1.1 System Purpose
- 1.2 Operational Scope
- 1.3 Document Reading Guidelines

2. Core Business Strategy & Expected Value

- 2.1 Two-Phase Implementation Plan
- 2.2 Measurable Expected Business Benefits

3. System Actors & Operational Roles Matrix

4. Visual Architecture & Graphical Design Models

- 4.1 UML Use Case Diagram
- 4.2 System Architecture N-Tier Block Diagram
- 4.3 Database Relational Schema (ER Diagram)

5. Functional Breakdown of Intelligent AI Modules

- 5.1 Module AI-1: Predictive Demand Forecasting
- 5.2 Module AI-2: Smart Reorder & Procurement Strategy Optimizer
- 5.3 Module AI-3: Product Placement Velocity Sorter
- 5.4 Module AI-4: Customer Payment Behavior Credit Risk Evaluator

6. System Interface Layouts & Wireframes

- 6.1 Portal Secure Login Screen
- 6.2 Core Inventory Status Dashboard
- 6.3 B2B Customer Order Management Panel
- 6.4 Intelligent AI Control & Insights Center

7. Non-Functional System Guarantees & Technical Parameters

1. INTRODUCTION & EXECUTIVE SUMMARY

1.1 System Purpose

This Software Requirements Specification (SRS) establishes a transparent operational blueprint for a smart Warehouse Management System (WMS). Designed primarily for wholesale B2B grocery networks, this system eliminates inventory visibility gaps, protects tracking states for fresh products, reduces expired waste, and introduces predictive dashboards to turn typical logistics backlogs into competitive advantages.

1.2 Operational Scope

The system connects international purchase management, internal warehouse barcode and QR tracking, batch allocation rules, wholesale client checkout workflows, and field team delivery updates. Secure web interfaces handle supplier and client administration, while crisp mobile interfaces empower warehouse floor operators and field dispatch riders.

1.3 Document Reading Guidelines

This guide is optimized for both corporate business clients and software technical teams. Business managers looking for concrete returns on investment can focus on Section 2.2, while engineering teams can use Section 4 and Section 5 as development blueprints.

2. CORE BUSINESS STRATEGY & EXPECTED VALUE

2.1 Two-Phase Implementation Plan

To reduce timeline risks and deliver immediate business utility, implementation features are split into a clear, two-stage track. High-importance components and core business logic are completed entirely during Phase 1.

Phase 1: Core System & Key AI Implementation Phase

- Secure User Management with explicit Role-Based Access Control (RBAC).
- Supplier Portal, International Purchase Order Tracking, and Inbound Stock Verification.

- Batch-Wise Inventory Management, Expiry Control, and Automated Low-Stock Notifications.
- B2B Wholesale Ordering Portal, Ledger Syncing, and Live Dispatch Status Trackers.
- Mobile Rider Dispatching, Route Progress Tracking, and Digital Proof of Delivery.
- **Core Practical AI Modules:** Predictive Demand Planning, Smart Reorder Recommendations, Inventory Velocity Sorting, and Customer Payment Risk Analysis.

Phase 2: Future Scalability Roadmap

- Integration with hardware automation extensions such as automated physical warehouse smart devices and RFID scanning grids.
- AI Route Optimization updates factoring in unexpected local road traffic delays.
- AI-Based Image Inspection updates to parse incoming bulk shipping pallets using automated optical camera verification feeds.
- Multi-country operational localized parameters and multi-warehouse currency tracking updates.

2.2 Measurable Expected Business Benefits

The implementation of this intelligence-driven WMS platform directly drives the following measurable business improvements:

TARGET PAIN POINT	ENFORCED SYSTEM BUSINESS BENEFIT
Stock Shortages & Out-of-Stocks	Reduces stockouts by cross-referencing live supply levels with dynamic procurement look-a-heads.
Perishable Product Expiry Waste	Minimizes material loss through proactive alert notifications sent 90 days before a batch expires.
Order Selection Discrepancies	Improves pick-and-pack fulfillment precision by validating item barcodes continuously on mobile devices.
Delivery Inefficiencies	Speeds up shipping turnaround times by routing clear, structured drop-off checklists to riders.
Manual Administrative Overheads	Replaces continuous data entry with auto-generated recommendation drafts for restocking plans.
Cash Flow Risks & Late Payments	Protects operational revenue cycles by automatically holding high-risk accounts when limits are breached.

3. SYSTEM ACTORS & OPERATIONAL ROLES MATRIX

The application restricts dashboard features through verified roles to protect data safety across the business network:

USER ACTOR ROLE	ALLOWED ACTIONS & PLATFORM OPERATIONAL SCOPE
Super Admin	Complete operational control, user account provisioning, system logs audit review, and global system parameter setup.
Supplier	Maintains product descriptions, details available batch volumes, and posts international transit updates.
Business Customer	Accesses wholesale pricing tables, creates bulk supply reservations, monitors balances, and checks delivery status.
Warehouse Staff	Verifies inbound containers, scans barcodes/QR labels, processes picking sheets, and updates loading statuses.
Sales Person	Onboards new wholesale clients, manages custom credit boundaries, and tracks regional product category demand.
Delivery Rider	Uses the mobile interface to follow destination routes, log delays, and upload customer signatures as Proof of Delivery.

4. VISUAL ARCHITECTURE & GRAPHICAL DESIGN MODELS

4.1 UML Use Case Diagram

This functional diagram visualizes the boundaries of your warehouse business system, illustrating how different operational staff and external entities engage with the central processing modules.

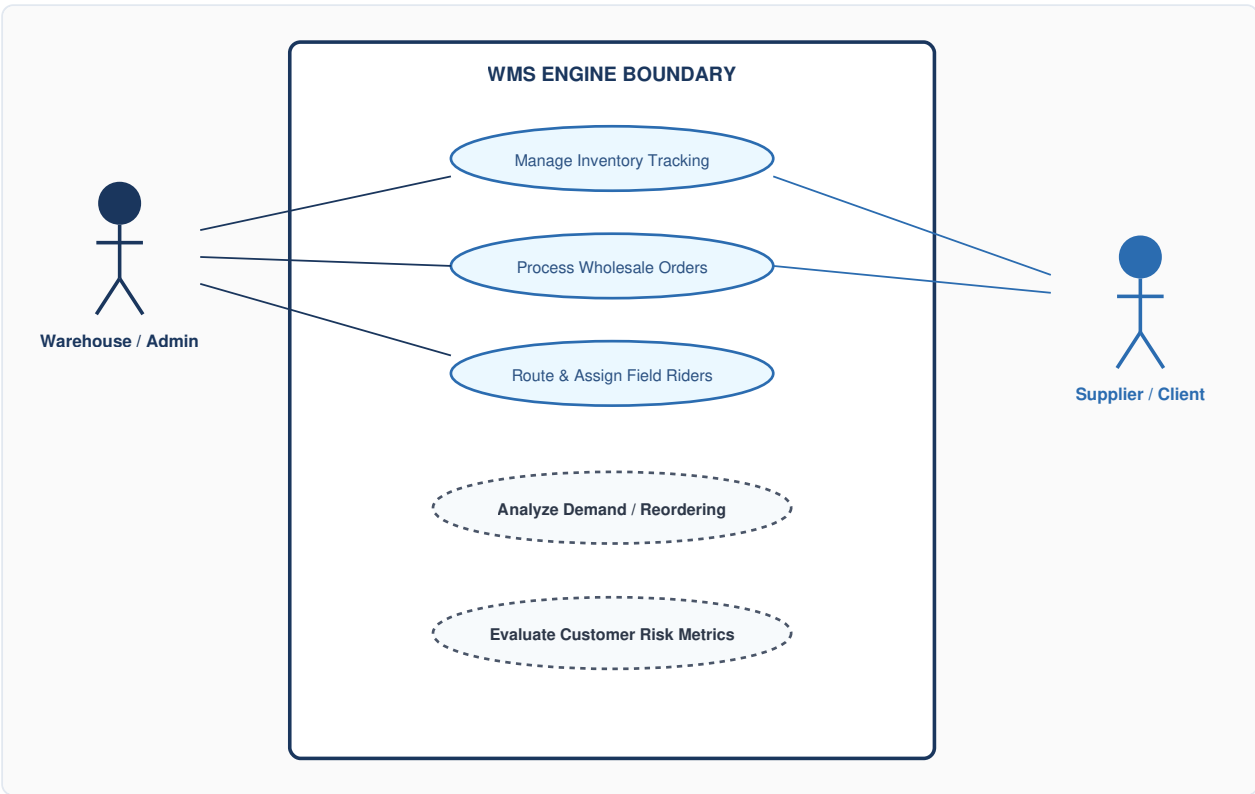


Figure 4.1: Technical System Boundary Use Case Map

4.2 System Architecture N-Tier Block Diagram

This technical view details the isolated layout of the solution, ensuring front-end web apps communicate securely into the main business database instance.

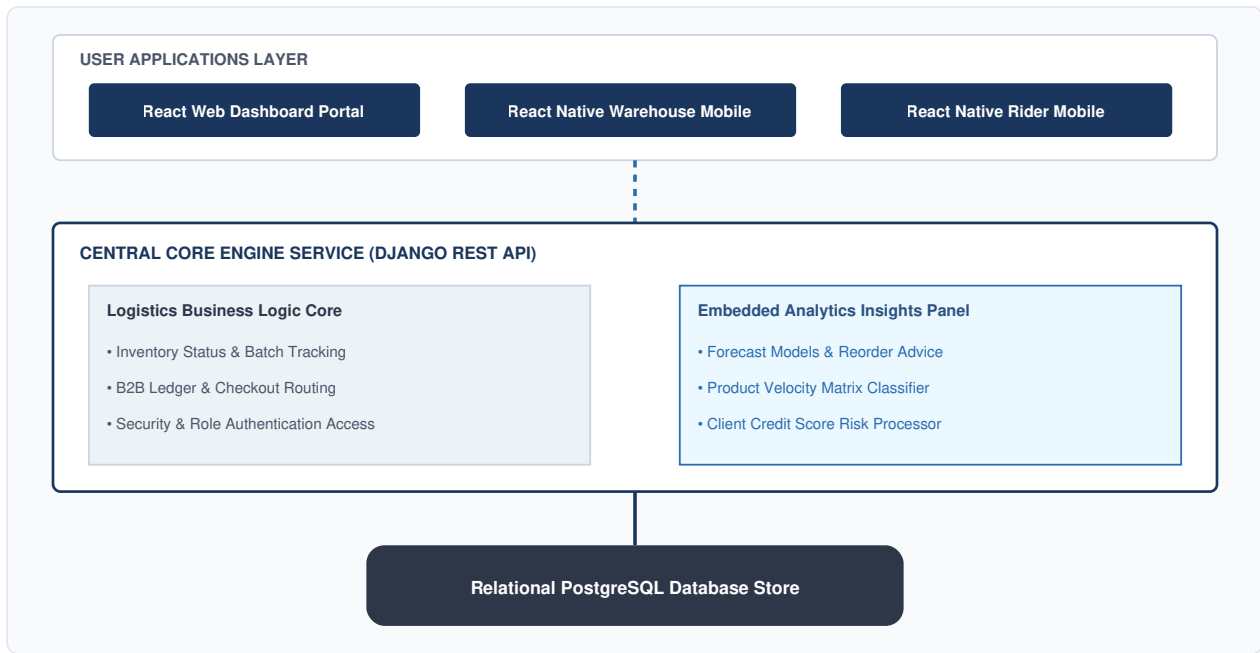


Figure 4.2: High-Fidelity N-Tier Solution Framework Layout

4.3 Database Relational Schema (ER Diagram)

This structural schematic presents the specific relational design and primary table connections powering the system backend database.

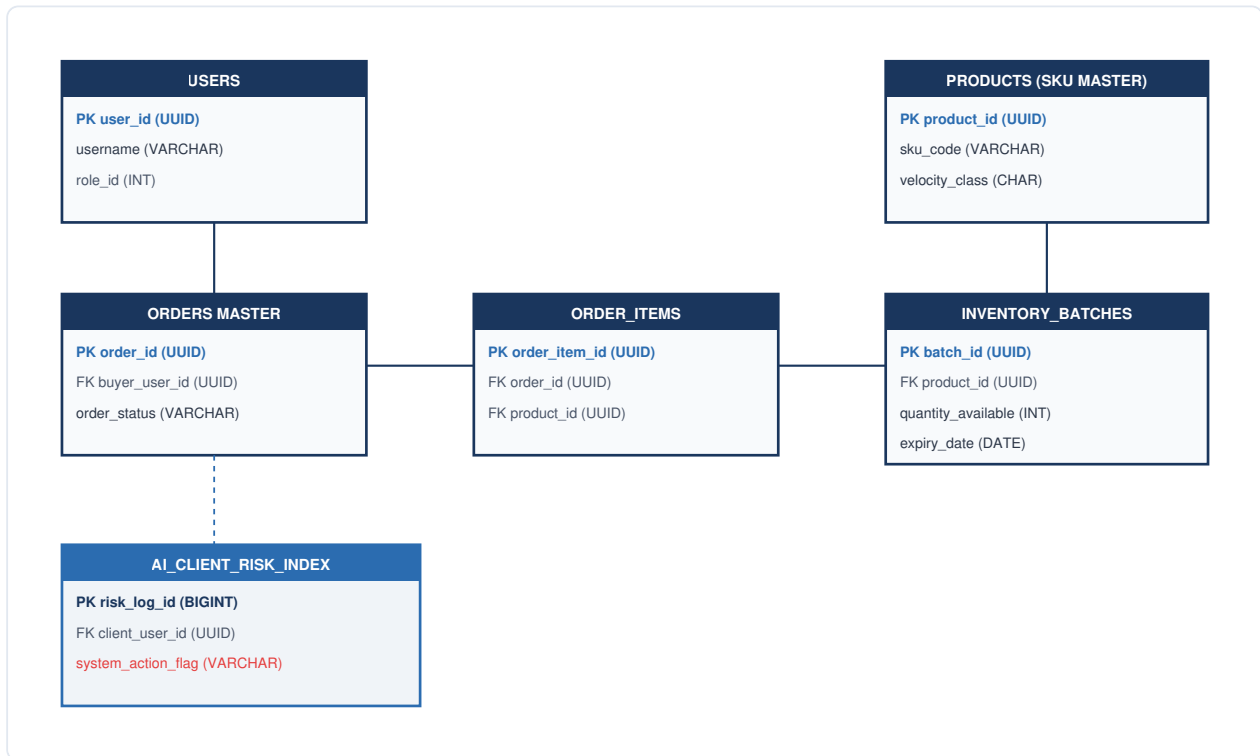


Figure 4.3: Relational Schema Logical Database Design Structure

5. FUNCTIONAL BREAKDOWN OF INTELLIGENT AI MODULES

To deliver advanced business analytics safely, the system utilizes integrated decision-support workflows. Instead of fully autonomous actions that could cause financial risks, the system acts as an expert assistant, generating clear optimization recommendations that always await manual management approval before execution.

5.1 Module AI-1: Predictive Demand Forecasting

5.1.1 Operational Business Logic & Processing Workflow

This module identifies upcoming inventory consumption cycles to protect stock continuity without over-allocating storage space. The system analyzes long-term sales patterns, localized wholesale order frequencies, and repeating monthly grocery delivery surges.

5.1.2 Data Source Dependencies

- Rolling 12-month transaction tracking ledger detailing historical volume sales quantities per SKU.
- Calendar dimension attributes outlining standard holiday schedules and wholesale seasonal milestones.

5.1.3 Downstream Application Workflows

The analytics layer processes these signals to calculate clear 30-day looking-forward quantity estimates for every stock category, updating the main system planner without relying on unreviewed minimum safety lines.

5.2 Module AI-2: Smart Reorder & Procurement Strategy Optimizer

5.2.1 Operational Business Logic & Processing Workflow

This engine acts as an electronic sourcing adviser, identifying the precise timing and quantity needed for inventory restocking. It evaluates ongoing warehouse storage holding constraints against single-run shipment costs to protect operational working capital cycles.

5.2.2 Data Source Dependencies

- Predictive consumption estimates delivered directly by Module AI-1.

- Active stock batch tracking tables mapping active shelf-life limits and expiration milestones.
- Supplier service records identifying recorded delivery turnarounds and bulk commercial pricing scales.

5.2.3 Downstream Application Workflows

When stock levels reach a critical line, the module generates structured purchase order recommendations for administrative approval. Drafts are held securely inside the master admin view, allowing a supervisor to sign off or update parameters with a single click before the order goes to suppliers.

5.3 Module AI-3: Product Placement Velocity Sorter

5.3.1 Operational Business Logic & Processing Workflow

To maximize physical warehouse efficiency, this engine classifies products dynamically based on transaction velocity footprints. Stock is automatically assigned to three specific categories based on historical picking frequencies:

- **Fast-Moving Stock:** High-volume items requiring rapid daily picking turnarounds.
- **Medium-Moving Stock:** Standard items showing reliable, steady transactional intervals.
- **Slow-Moving Stock:** Low-frequency items matching seasonal storage needs.

5.3.2 Data Source Dependencies

- 90-day aggregate warehouse stock selection tracking maps and inventory turnover logs.
- Warehouse coordinate matrix detailing precise floor walking distances to primary outbound packing zones.

5.3.3 Downstream Application Workflows

The core inventory router flags these status updates to direct inbound floor workers. Newly received fast-moving items are systematically guided to floor levels adjacent to loading bays to minimize picking times, while slow-moving items are assigned to upper racking positions.

5.4 Module AI-4: Customer Payment Behavior Credit Risk Evaluator

5.4.1 Operational Business Logic & Processing Workflow

This financial safety tool evaluates client exposure limits before orders advance to the warehouse floor picking queue. By cross-referencing invoice completion milestones against credit balances, the system flags higher-risk transactions to protect business cash flow.

5.4.2 Data Source Dependencies

- Historical client payment ledgers detailing delayed invoice tracking trends.
- Corporate credit allocation registers outlining current outstanding balances vs. absolute limits.

5.4.3 Downstream Application Workflows & Hard-Coded Decision Rules

The core order engine screens incoming wholesale checkouts against these risk categories, systematically enforcing the following concrete operational rules:

ASSIGNED STATUS CATEGORY	ENFORCED AUTOMATIONS & SYSTEM CONSTRAINTS
Low Operational Risk	The system automatically signs off on transaction approval, confirms physical batch reservation, and moves picking workflows directly to floor staff tablets.
Moderate Risk Boundary	The transaction is paused under a soft managerial warning flag. The system extends a strict 7-day payment grace window and alerts the assigned Sales Person to verify account records.
Critical Risk Alert	The system triggers a hard administrative credit hold. Stock allocation is instantly denied to protect physical inventory assets, and order execution is blocked until outstanding balances are cleared.

6. SYSTEM INTERFACE LAYOUTS & WIREFRAMES

To ensure alignment before backend engineering begins, this section details the user experience wireframe structures designed for the web portals and mobile clients.

6.1 Portal Secure Login Screen

Web Secure Authentication Interface - Secure Login

Enterprise Gateway Access

Username / Email Input

admin@enterprise-wms.com

Password Security Input

.....

Assigned Role Boundary

☒ Admin

☐ Supplier

☐ Client

☐ Sales

SUBMIT ENCRYPTED SIGN-IN

6.2 Core Inventory Status Dashboard

Web Portal Dashboard - Real-Time Inventory Control

Quick Operational Status Summary: Total SKUs: [1,240] | Low Stock Alerts: [14] | Expiring Batches (90d): [6]

SKU CODE	PRODUCT DESCRIPTION	AVAILABLE STOCK	BATCH CODE	EXPIRY STATUS
SKU-GRO-4022	Organic Wholesale Whole Eggs	12,400 Units	BAT-2026-X9	Stable (180 Days Remain)
SKU-GRO-1105	Premium Fresh Chicken Breast	1,150 Units	BAT-2026-A2	CRITICAL DISPATCH (8 Days)

TRIGGER SYSTEM RE-SCAN

DOWNLOAD EXCEL EXPORT

6.3 B2B Customer Order Management Panel

Client Dashboard Portal - Wholesale Checkout Pipeline

Customer Profile Link: Metro Distributing Corp | **Open Available Credit:** \$45,000 / \$50,000

Active Basket SKU Allocations:

- SKU-GRO-4022 (Wholesale Eggs) x 200 Cartons → [Batch Locked]
- SKU-GRO-8812 (Long Grain Rice) x 150 Sacks → [Batch Locked]

Current Order Pipeline Progress Tracking State:

1. Pending Payment → 2. Batch Allocation → 3. Warehouse Pick → 4. Shipping Transit

COMMIT WHOLESALE REQUISITION

6.4 Intelligent AI Control & Insights Center

Management Console - Business Intelligence Dashboard

Automated Procurement Plan Recommendation:

Advised Action: Purchase 500 units of SKU-GRO-4022 from Supplier A within 48 Hours.

Business Justification: Monthly regional trend data metrics show a 22% marketplace consumption spike arriving next period. Supplier delivery turnaround timelines align perfectly with this safety target window.

GENERATE REORDER PURCHASE DRAFT

Automated Financial Safety Flags:

Order #10944 (Global Food Group): **CRITICAL SAFETY HOLD SECURED**. Customer risk index reaches 0.88 due to consistent outstanding settlement invoice tracking backlogs. Warehouse allocation tasks paused systematically awaiting asset verification.

7. NON-FUNCTIONAL SYSTEM GUARANTEES & TECHNICAL PARAMETERS

- **NFR-1.1 (System Performance Response):** The central interface network handling core API operations must complete simple operational queries in less than 2.0 seconds during standard use spikes.
- **NFR-1.2 (Ledger Concurrency Safeties):** Database record allocation parameters must use atomic blocking modes to guarantee duplicate reservations are blocked when multiple bulk customers access identical batch balances simultaneously.
- **NFR-2.1 (Transport Security Framework):** Every external communication link and transactional query path must be restricted to standard TLS 1.3 encryption protocols.
- **NFR-2.2 (System Change Audits):** Every structural change affecting product catalog details, client credit flags, or quantity modifications must instantly execute a read-only record write to an isolated database audit ledger.
- **NFR-3.1 (Mobile Network Loss Resilience):** Field driver mobile configurations must log package status events in secure device memory tables if regional cell tower connection drops, automatically sync-shipping updates back to the primary server database once network signal returns.